



Data Science Foundations

Outline

- Introduction to Data Science
- Data Science Lifecycle
- Roles
- Data Science vs. AI

Acknowledgement : some slides are enhanced using AI particularly CoPilot and Gemini

What Is Data Science? 🔍

- **Definition:** collecting, analyzing, and modeling data to **generate insights** (Data Storytelling) that drive **decision-making**

 **Data** →  **Insight** →  **Decision**

 **The goal:** better decisions, not pretty charts

- Learning about the world from data using computation
- Not just reporting: asks new questions, not just old ones
- **Interdisciplinary:** math, programming, machine learning, domain knowledge
- **Structured and unstructured data:** tables, text, images, audio, sensors data.

Difference Between Data, Information, and Insights

Level	Example
Data	Sales = \$100,000 in October, \$130,000 in November
Information	Sales increased by 30% from October to November
Insight	Holiday shopping behavior drives demand, creating an opportunity for seasonal promotions

A good insight is:

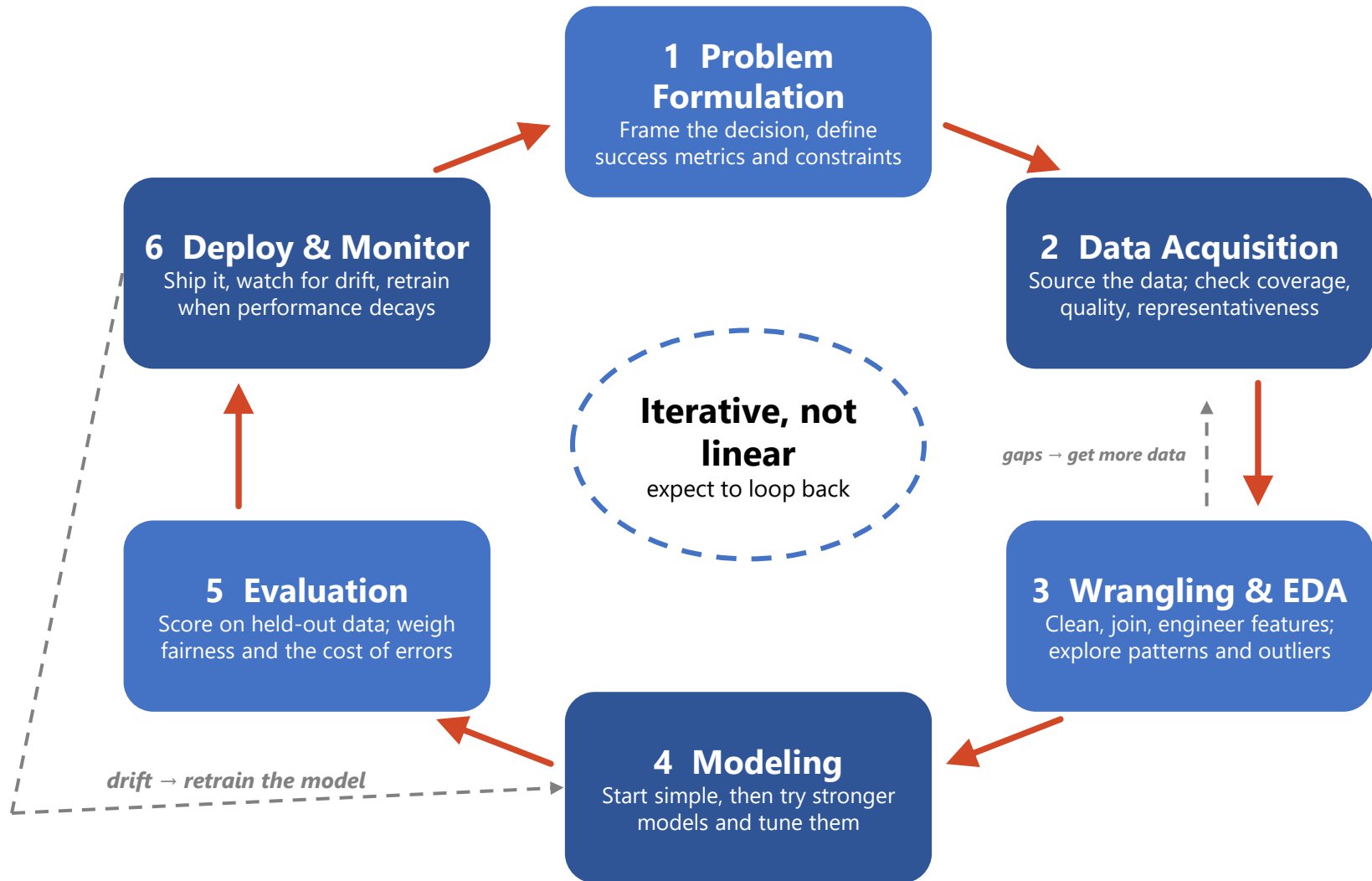
- **Relevant** to the business or research question
- **Actionable** and useful for decision-making
- **Evidence-based** and supported by data
- **Non-obvious**, revealing something not immediately apparent
- **Impactful**, leading to improved outcomes

Where Data Science Creates Value

-  **Healthcare:** imaging diagnosis, readmission risk, drug discovery
-  **Finance:** fraud detection, credit scoring, risk management
-  **Retail:** demand forecasting, recommenders, dynamic pricing
-  **Logistics:** route optimization, predictive maintenance, inventory
-  **Education:** admissions analytics, early warning for at-risk students
-  **Energy:** demand forecasting, policy evaluation

Data Science Lifecycle

Data Science Lifecycle



Types of Data Analysis

-  **Descriptive** — “What happened?” Reporting on past data with KPIs and dashboards.
Example: Dashboard of student applications, enrollment and pass rates by program.
-  **Diagnostic** — “Why did it happen?” Drill-down and comparison to find root causes.
Example: Enrollment fell 15% is associated to fewer international applications and lower CE admissions.
-  **Predictive** — “What is likely to happen?” Statistical and ML prediction models on historical data.
Example: Predicting first-year students at risk of failing based on early grades, attendance, and engagement.
-  **Prescriptive** — “What should we do?” Uses optimization, simulation, and decision models to recommend actions that achieve desired outcomes under constraints.
Example: Splitting a fixed scholarship and tutoring budget across students to minimize predicted student dropouts.
-  **Cognitive** — “What does the data mean, and how can AI interact with it?”
Uses NLP, LLMs, and multimodal AI to interpret unstructured information, generate insights, answer questions, and support or automate knowledge-based tasks.
Example: a tutor that answers questions from the course material with citations, grades short answers against the rubric, and reports recurring misconceptions.

 **Rising difficulty, rising value:** each level builds on the one before it.

What we will focus on in this course

- **Data Acquisition and Wrangling — getting trustworthy data**
 - Sourcing, data quality, and representativeness
 - Cleaning, transformation, integration, feature engineering
- **Exploration — finding patterns and structure**
 - Exploratory data analysis (EDA) and visualization
 - Clustering and dimensionality reduction (unsupervised)
- **Inference — drawing reliable conclusions**
 - Probability, sampling, and uncertainty
 - Statistical testing, estimation, and A/B testing
 - Causal inference and experimental design
- **Prediction — modeling to forecast outcomes**
 - Regression and classification
 - Model evaluation, and validation

Roles

Who Does What: Roles Across the Data Science Lifecycle

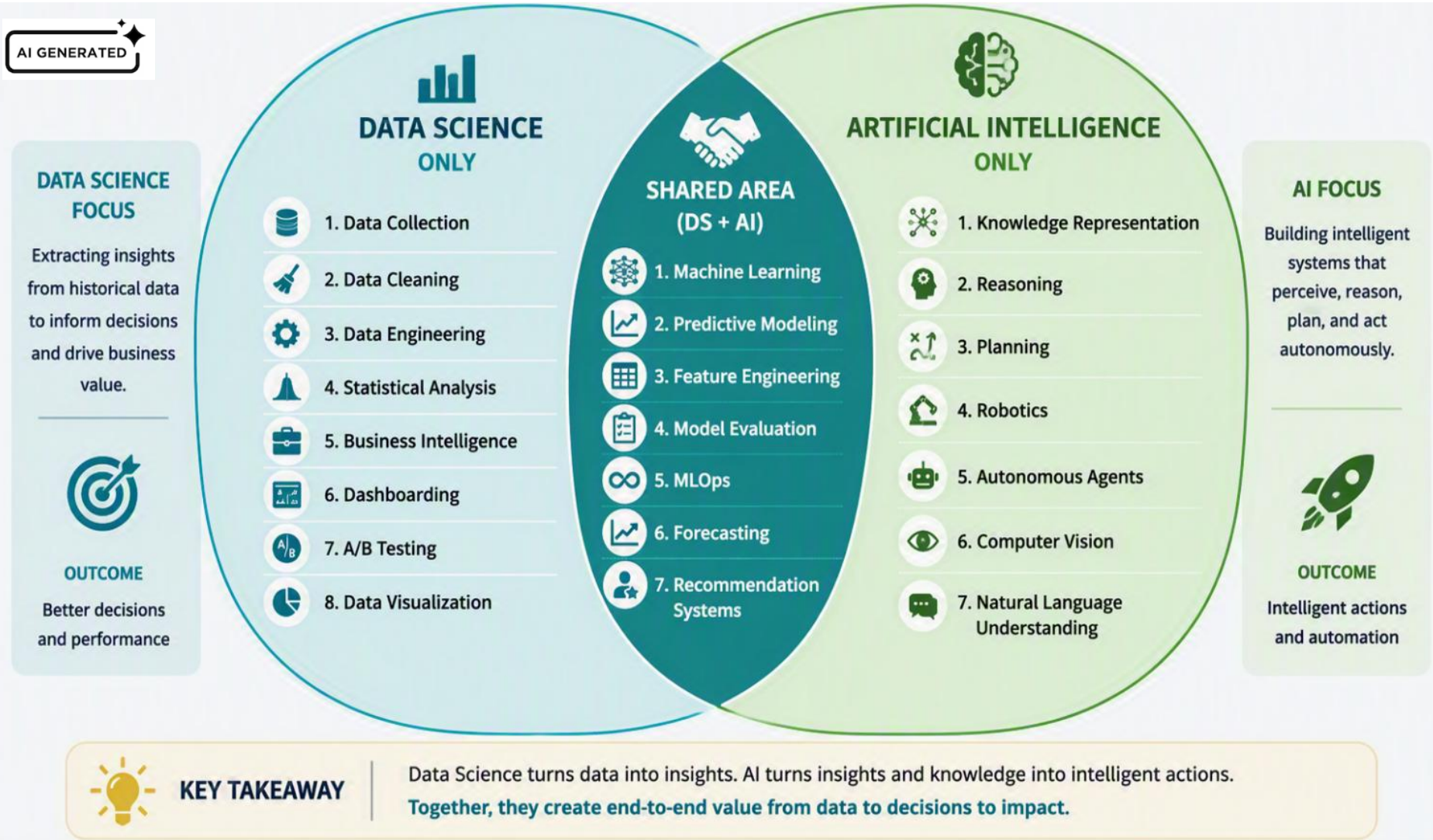
Role	The question they answer	What they deliver	Typical tools
 Data Engineer <i>Owens stage 2, enables stage 3</i>	<i>"Is the data complete, correct and on time?"</i> Role: Moves data from source systems into the warehouse and keeps it reliable.	Data pipelines, clean tables, freshness and quality checks	SQL, Spark, Airflow, cloud data platforms
 Data Analyst <i>Owens stage 3, informs stage 1</i>	<i>"What happened, and what is driving the numbers?"</i> Role: Turns a business question into a metric and explains why it moved.	Metric definitions, dashboards, trend and segment analysis	SQL, BI tools, spreadsheets, visualization
 Data Scientist <i>Owens stages 1, 4 and 5, shares 3</i>	<i>"What causes it, and what happens next?"</i> Role: Frames the problem, tests hypotheses, builds and validates stats/ML models.	Framed problem, A/B tests, validated stats/ML models	Python/R, statistics, A/B testing, ML algorithms
 ML Engineer <i>Owens stage 6, supports stage 4</i>	<i>"Will the model run reliably and fast in production?"</i> Role: Deploys the model as a service, monitors accuracy and speed, retrains or rolls back when it slips.	Model serving APIs, monitoring and alerts, retraining and rollback	MLOps, Docker, CI/CD, model registries
 AI Engineer <i>Owens stage 6 for LLM based apps, evaluates in 5</i>	<i>"Can a foundation model do this accurately and safely?"</i> Role: Builds LLM apps grounded in company data and measures accuracy, cost and risk.	LLM apps, retrieval grounding, guardrails, evaluations	Prompting, RAG, agents, evaluation

*The analyst's finding becomes the scientist's question; the scientist's model becomes the ML engineer's service.
In a small team, one person wears several hats.*

Data Science vs. AI

AI and DS: Overlaps and Unique Characteristics

Two disciplines with shared foundations and distinct strengths



DS extracts insights from data. AI uses those insights to create intelligent systems

AI GENERATED

WHAT DATA SCIENCE EXCELS AT



Extracting value and insights from data.

Key Outcomes

- ✓ Deeper understanding of data
- ✓ Data-driven insights
- ✓ Performance tracking
- ✓ Better business decisions

DATA SCIENCE

Focus: Understanding Data



- 1. Data Collection**
Gather raw data from various sources



- 2. Data Cleaning**
Handle missing values, remove noise, fix errors



- 3. Data Engineering**
Build pipelines, transform and prepare data



- 4. Statistical Analysis**
Summarize, test hypotheses, find patterns



- 5. Business Intelligence**
Turn data into actionable business insights



- 6. Dashboarding**
Monitor KPIs and performance through dashboards



- 7. A/B Testing**
Run experiments to validate ideas and decisions



- 8. Data Visualization**
Communicate insights clearly and effectively



SHARED AI + DATA SCIENCE



- 1. Machine Learning**
Algorithms that learn from data



- 2. Predictive Modeling**
Predict outcomes and behaviors



- 3. Feature Engineering**
Create informative features



- 4. Model Evaluation**
Assess performance and accuracy



- 5. MLOps**
Deploy, monitor and improve models



- 6. Forecasting**
Predict future trends and values



- 7. Recommendation Systems**
Suggest relevant items or actions

ARTIFICIAL INTELLIGENCE

Focus: Intelligent Action



- 1. Knowledge Representation**
Structure knowledge for machines to use



- 2. Reasoning**
Draw conclusions from known facts



- 3. Planning**
Make plans to achieve goals



- 4. Robotics**
Perceive and interact with the physical world



- 5. Autonomous Agents**
Operate independently to achieve objectives



- 6. Computer Vision**
Understand and interpret visual information



- 7. Natural Language Understanding (NLU)**
Understand and generate human language

WHAT AI EXCELS AT



Using insights and knowledge to perceive, reason, decide, and act autonomously.

Key Outcomes

- ✓ Automation
- ✓ Intelligent decisions
- ✓ Real-world interaction
- ✓ Adaptation and learning in dynamic environments

FROM DATA TO INTELLIGENCE: THE CONTINUUM



- 1. Data Collection**
Raw data from multiple sources



- 2. Data Preparation**
Clean, transform and integrate



- 3. Exploratory Analysis**
Discover patterns and insights



- 4. Machine Learning**
Train models to make predictions



- 5. Deep Learning**
Use neural networks for complex tasks



- 6. Generative AI**
Create new content, ideas and solutions



- 7. Intelligent Systems**
Deploy to solve real-world problems at scale



KEY TAKEAWAY

Data Science and AI are complementary disciplines. Data Science uncovers insights from data. AI leverages those insights to build intelligent systems that perceive, reason, decide, and act. Machine Learning is the vital intersection that connects both worlds.



Different Strengths



Shared Methods

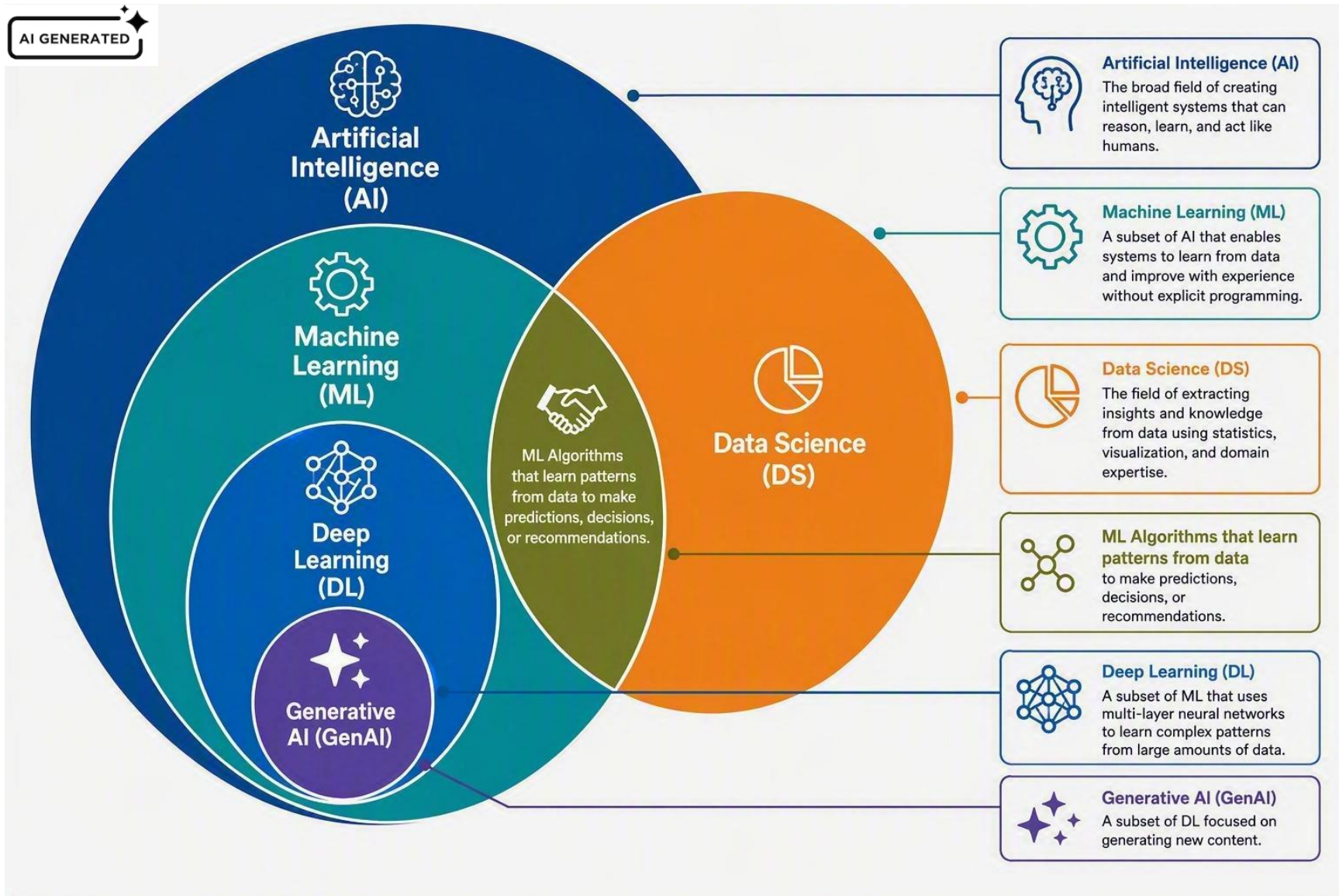


Common Goal: Impact

How DS, ML, DL, and Gen AI Relate

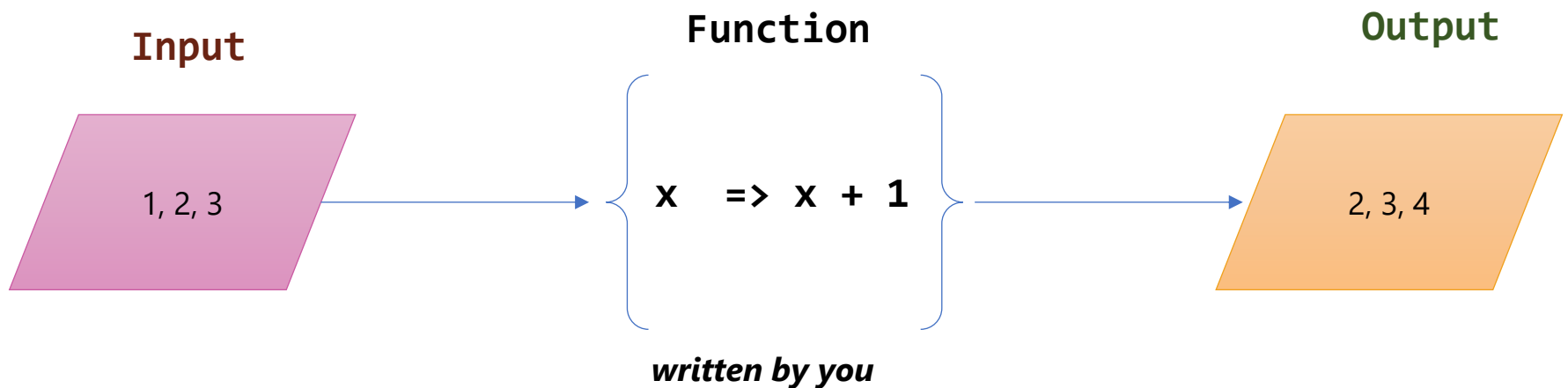
- **Data Science:** Extracts insights, predictions, and business value from data using statistics, computing, and domain knowledge.
- **AI:** Builds systems that perceive, reason, learn, and act to accomplish tasks that normally require human intelligence.
- **ML (Intersection):** Algorithms that learn patterns from data to make predictions, decisions, or recommendations.
 - **Deep Learning:** ML approach using neural networks to learn complex representations from large-scale data.
 - **Generative AI:** Models that create new content such as text, images, audio, video, and code.

Data Science vs. AI



Coding : Input + Function → Output

- You already know the mapping from input to output, so you **write the Function** yourself — the computer just applies it to every Input
- In **ML the mapping is unknown**: you supply Input and Output, and the model learns the Function

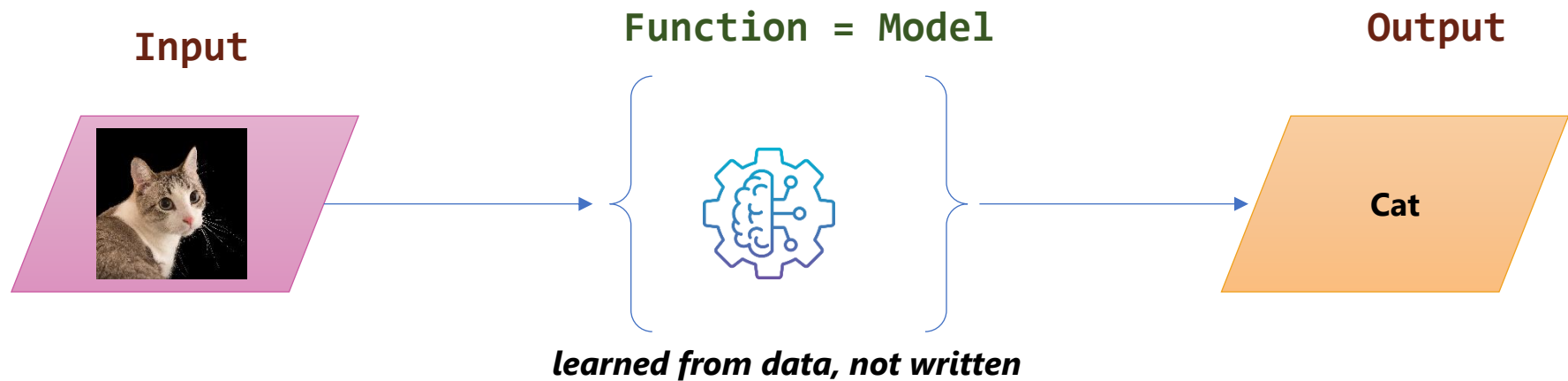


Coding: the function is known, the outputs are computed

Model building: the outputs are known, the function is learned

ML : Input + Output → Function (Model)

- No one can write a function for “is this a cat?” — so you collect thousands of **labelled examples**: an Input with its known Output
- Training finds the function that best **approximates** that mapping
 - The learned **function** is what we call the **model**. It can then be used to classify images it has never seen before



Training: many (image, label) pairs → the model

Prediction: a new image → the trained model → its label

Types of ML

AI GENERATED



MACHINE LEARNING

Algorithms that learn patterns from data to make predictions or decisions.

[Link to source](#)



SUPERVISED LEARNING

Learn from labelled data

Goal:

Learn a mapping from inputs to known outputs.



UNSUPERVISED LEARNING

Learn from unlabelled data

Goal:

Discover hidden patterns or structures in the data.



REINFORCEMENT LEARNING

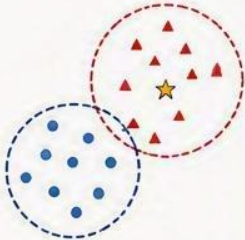
Learn by interacting with an environment

Goal:

Maximize cumulative reward through trial and error.

CLASSIFICATION

Predict a discrete class or category.

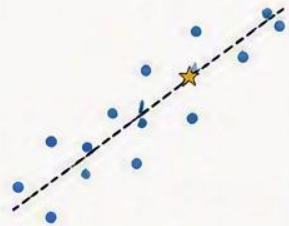


Examples:

- Spam detection
- Image classification
- Disease diagnosis

REGRESSION

Predict a continuous numeric value.

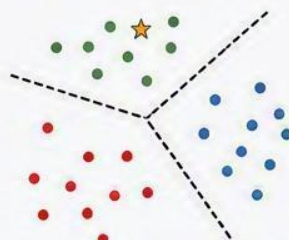


Examples:

- House price prediction
- Sales forecasting
- Temperature prediction

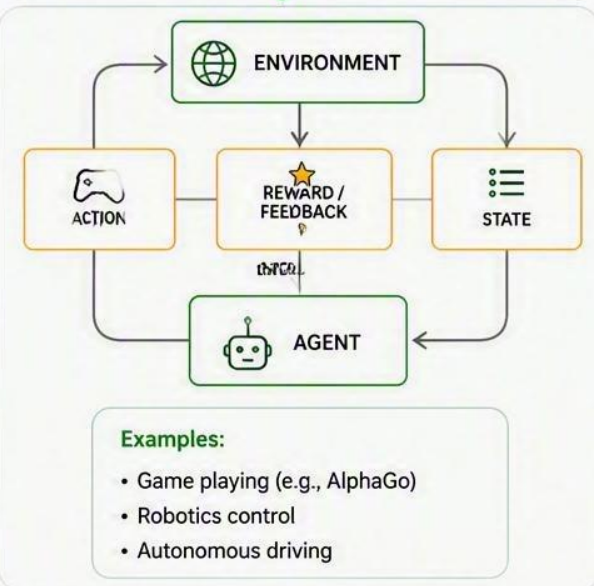
CLUSTERING

Group similar data points together.

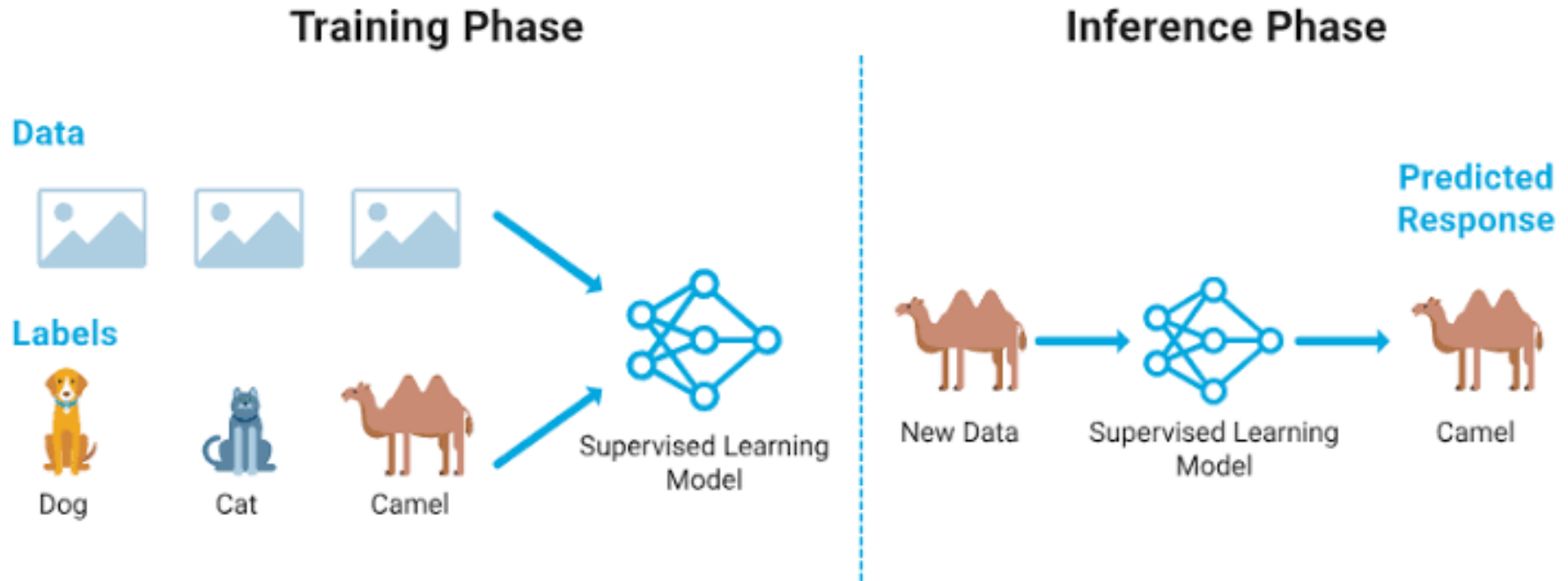


Examples:

- Customer segmentation
- Document grouping
- Anomaly detection



Supervised Learning – Classification Example



- **Inference:** apply the **trained prediction function** to a new image to predict its class

Supervised Learning – Regression Example

Training Phase

Features



350 sq m,
5 bed, 6 bath



Location:
Lusail



Year Built:
2010

QR 4,500,000



QR 3,800,000



QR 5,100,000



Target (Prices)



Regression Model
(Trained)

Inference Phase

New Data

New Features



New House Data:
380 sq m, 6 bed, 7 bath
Location: The Pearl-Qatar (New Location)
Smart Home features, Floorplan B



Regression Model
(Trained)

Predicted Response

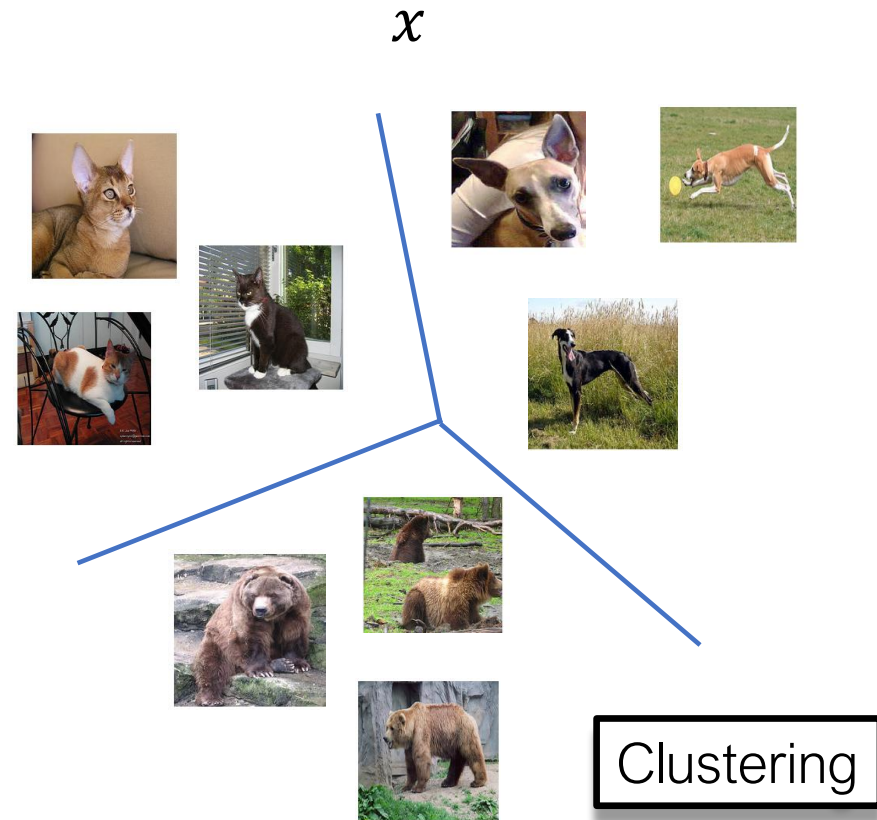
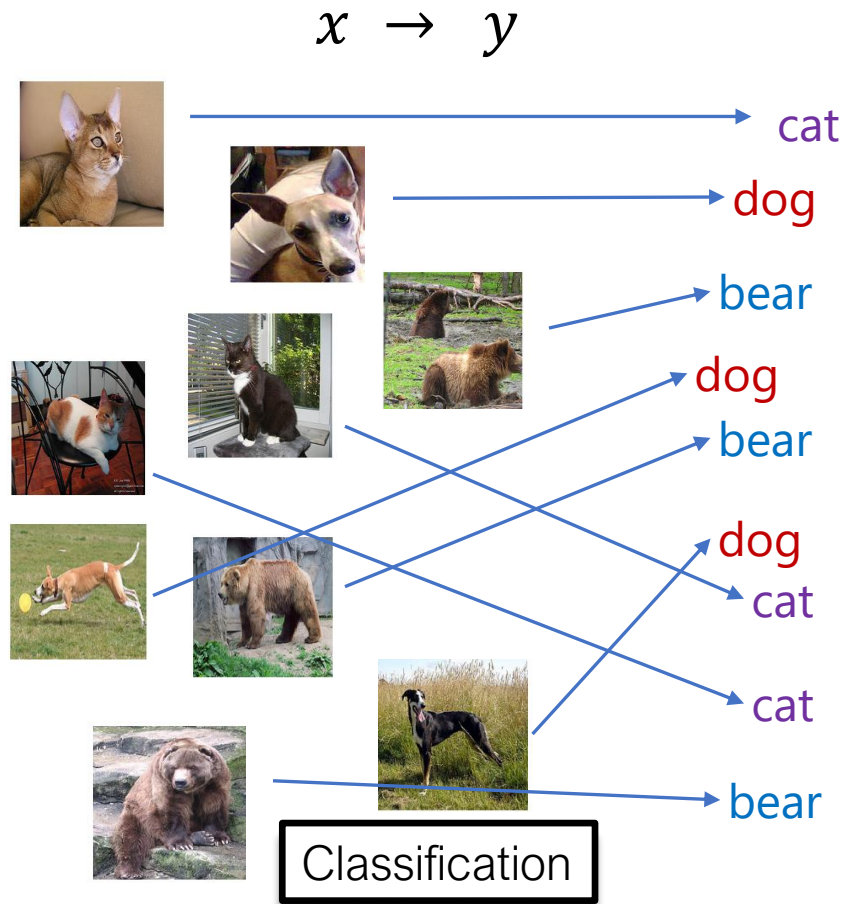


\$4,4750,000

Estimated
Value

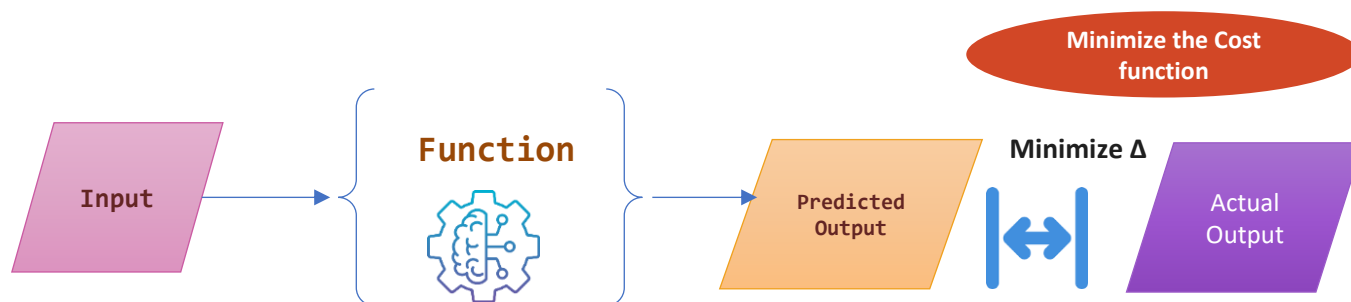
- **Inference:** Apply the trained **regression function** to the house's features (such as size, location, and year built) to predict its estimated value.

Supervised Learning vs Unsupervised Learning



ML: learn a **Function** that minimizes the cost

- **Guess** — start with random parameters, so the first predictions are wrong
- **Measure** — the cost function scores how far the Predicted Output sits from the Actual Output
 - also called the loss or error function
- **Adjust** — nudge the parameters to lower the cost, then repeat until it stops improving



Example: predicting house prices

- The model: **Price = $w \times \text{Size} + b$**
 - w and b are the two numbers to learn from past sales
- **Guess** — start at $w = 0$, $b = 0$: the line is flat and the error is huge
- **Measure** — the Mean Squared Error (MSE) scores the line: the average squared gap between predicted and actual price
- **Adjust** — tilt it with w and shift it with b so the MSE drops, then repeat
- Result: **$w \approx \text{QR } 12,000 \text{ per sqm}$, $b \approx \text{QR } 150,000$** — the MSE stops falling, so the line has settled at $\text{Price} \approx 12,000 \times \text{Size} + 150,000$



Blue dots = past sales | Red line = the model

w (slope) — each extra sqm adds about QR 12,000

b (intercept) — the baseline QR 150k before size counts

Cost function — Mean Squared Error: $\text{MSE}(w, b) = (1/n) \sum (\text{Actual price} - \text{Predicted price})^2$

Training is the search for the w and b that make the MSE smallest.

Resources

- **Foundations & data wrangling** — [Python Data Science Handbook](#)
- **Exploratory analysis & visualization** — [Fundamentals of Data Visualization](#)
- **Probability, sampling & uncertainty** — [Seeing Theory](#) (Brown University)
- **Machine learning, hands-on** — [Google Machine Learning Crash Course](#)
- **Clustering & dimensionality reduction** — [scikit-learn User Guide](#)
- **Statistical testing & A/B experiments** — [Microsoft Experimentation Platform](#)
- **Causal inference & experimental design** — [Causal Inference](#)